Algebra: Chapter 0 Solutions

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Preface

This document does not seek to represent the perfect solutions manual to the text at hand, and is more of a way of recording my progress through the text. Should you happen to find this, and wish to contribute or point out any logical flaws in the arguments presented, I would be more than happy to adjust my solutions to perhaps work towards establishing this as a suggested solutions. Should this end up being the case, I will update this section accordingly

1 Chapter 1: Preliminaries: Set Theory and Categories

1.1 Naive Set Theory

1.1.1 Sets

Exercise 1.1. Locate a discusiion of Russell's paradox, and understand it.

Solution There are many discussions of the paradox. The one I found most intuitive or easy to understand was the labelling of sets as such: We say a set is 'abnormal' if the set contains itself ($S \in S$), or we say that a set is 'normal' if it does not contain itself ($S \notin S$). We now define the set X as the set of all normal sets. We ask the question 'is X a normal or abnormal set?'

Suppose it is normal, then it would be contained within X by definition, making X abnormal. If we instead suppose X is abnormal, then X would not be contained in itself, making X normal.

We arrive at a paradox; that is, X is neither normal or abnormal.

□

Exercise 1.2. Prove that if $\sim$ is an equivalence relation on a set S, then the corresponding family $\mathscr{P}_\sim$ defined in §1.5 in indeed a partition of S: that is, its elements are nonempty, disjoint and their union is S.

Solution We show the desired result by considering the definition of the equivalence class of a given in the text

$$[a]_\sim := \{b \in S | b \sim a\}$$

Firstly, its clear that every element $s \in S$ is contained in its own equivalence class. This is due to the reflexitivity requirement of the relation ($s \sim s$). This also shows that every equivalence class is non-empty. We now show that for any two equivalence classes $[a]_\sim$ and $[b]_\sim$ are either disjoint or equal.

Suppose the two sets are not disjoint. Then there exists an element $z \in [a]_\sim \cap [b]_\sim$. Let $n \in [a]_\sim$ be arbitary. Since we have $n \sim z$ and $z \sim b$, we must have that $n \in [b]_\sim$ by transitivity and so $[a]_\sim \subseteq [b]_\sim$. Similarly, for an arbitrary element $m \in [b]_\sim$ we have $m \sim z$ and $z \sim a$ and so $m \in [a]_\sim$ implying that $[b]_\sim \subseteq [a]_\sim$. Combining these two results means that $[a]_\sim = [b]_\sim$ and the two equivalence classses are equal.

Totalling the results thus far, we see that all elements of S are in a equivalence class, and that these equivalence classes are either disjoint or equal, and so $\mathscr{P}_\sim$ indeed forms a partition of S.

□

Exercise 1.3. Given a partition $\mathscr{P}$ on a set S, show how to define a relation $\sim$ on S such that $\mathscr{P}$ is the corresponding partition.

Solution Let P_i denote the disjoint non empty subsets in $\mathscr{P}$. We define an equivalence relation on elements of S by the following: $a \sim b$ if and only if a and b are contained within the same subset P_i